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Body mass in comparative primatology

Data are presented on adult body mass for 230 of 249 primate species, based on a review of the literature and previously unpublished data. The issues involved in collecting data on adult body mass are discussed, including the definition of adults, the effects of habitat and pregnancy, the strategy for pooling data on single species from multiple studies, and use of an appropriate number of significant figures. An analysis of variability in body mass indicates that the coefficient of variation for body mass increases with increasing species mean mass. Evaluation of several previous body mass reviews reveals a number of shortcomings with data that have been used often in comparative studies.

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There are many research problems for which information is needed on sex-specific adult body mass of one or more primate species. The primary literature on body mass is vast, with data often published as incidental comments in studies concerned with other topics. Therefore, secondary sources with broad interspecific compilations of body mass have proven to be useful, such as those by Leutenegger & Cheverud (1982), Jungers (1985), Harvey & Clutton-Brock (1985), Fleagle (1988), and Ford & Davis (1992). However, no source represents an attempt to comprehensively survey the literature on body mass for all primate species. A number of critical decisions affecting the quality of body mass estimates have received little or no attention.

The purpose of this study is to review the issues involved in determining body mass for primate species and to present a new summary of these data.

Issues in data collection

The advantages and disadvantages of body mass as the basis for comparing species of different size have been discussed (e.g., Jungers, 1984*a*; Ford & Corruccini, 1985; Harvey & Krebs, 1990; Hartwig-Scherer & Martin, 1992; Smith, 1993), and will not be addressed here. It is assumed that an informed decision has been made to use body mass. The question is how to best select data once that decision has been made.

It is remarkable how little discussion has taken place concerning many of the issues involved in this process. At best, authors mention whether or not captive specimens were included, the criteria used to define adult status, whether or not pregnant and/or lactating females were included, and the sample size. The treatment of even these issues usually is superficial; statements may be made concerning each of the above, but the implications of these choices or the assumptions underlying them are not presented. The following is a review of issues that should be considered when evaluating body mass data from the literature.

Mass versus weight

A minor but ever-present issue of terminology—whether to use the term "mass" or "weight"—requires clarification before turning to substantive issues. In the literature, body weight and body mass are both commonly seen. There is no ambiguity at all, however, about

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which term is correct (Pennycuick, 1988, 1992). In the Système Internationale (SI), since 1960 the official protocol for units in science, kilograms are units of mass. The units for weight, which are identical to those of force, are newtons, which have dimensions of mass × length × time^{-2}. While it may seem that no one could be confused about the use of "body weight," data may be used in unforeseen contexts, and by workers with unforeseen backgrounds, including different native languages and different educational traditions concerning units of measurement. The units involved in calculations related to flight, work, fluid movement, metabolism, etc. can be complex (Schepartz, 1980), and Pennycuick (1988) described the confusion between systems of measurements as "a common source of disastrous error." Dimensional analysis is becoming increasingly important in ecology (e.g., Charnov, 1993), and because mass and weight have different dimensions, workers in dimensional analysis are careful to use the term mass for a quantity expressed in kilograms (Schepartz, 1980; McMahon & Bonner, 1983; Zlokarnik, 1991). There is, furthermore, no cost associated with use of the correct term.

Significant figures

There has been no serious treatment of the validity of the number of figures reported for primate body mass. Thus, no studies have examined measurement error for the purpose of validating the number of digits reported. The weighing apparatus is rarely described, and even more rarely is it acknowledged that a scale or balance has limited precision and accuracy.

Although it is common to see four significant figures reported for primate body mass, it is unlikely that the last digit is creditable in many circumstances. Animals weighing under 1·0 kg require only three digits for a description to the closest 1 g, and there are many scientific balances available that are accurate to 1 g or less for specimens under about 2 kg. However, for most larger animals, spring scales are used (e.g., Glander *et al.*, 1992). A high-quality spring scale has an accuracy (which cannot be assumed from the gradations on the scale) of about ± 0·3%, a 30 g range for a 5 kg animal (Forestry Suppliers, Inc., 1996). A third digit (the second decimal value) for a 5 kg animal implies a 10 g range, an accuracy of ± 0·1%. High-quality physician's strain gauge scales for weighing adult humans are accurate to ± 100 g. Thus, it is unlikely, in most circumstances, that the instruments used to determine body mass in the field allow for more than three valid significant figures. In addition, three figures is reasonable for data that must be converted from pounds and/or ounces to kilograms and grams.[1]

It appears that the use of four figures often follows directly from the decision to report mass in grams. It is common to see lists of body mass in which the last digit is always zero, for example, a value of 7530 g. This zero is a significant figure, indicating that mass is between 7529·5 and 7530·5 g. If a scale allows measurement to ± 10 g and data must be listed in grams, an exponential expression should be used, such as $75{\cdot}3 \times 10^2$ g. It is more convenient to use kilograms, and report the value as 7·53 kg.

The data presented in this study are therefore generally limited to three digits, even when four or more are reported in the cited study, and are presented in kilograms. Male gorillas, with a mean mass exceeding 100 kg, are reported to four figures, and species with mean masses of less than 0·10 kg (100 g) are reported to the nearest gram, resulting in two significant figures. Only two figures are reported for animals of any size when the source is limited to this precision.

[1]In this study, data reported as pounds and ounces in the original literature were converted as follows: kg=lb × 0·4536, and g=oz × 28·35.

The consideration of significant figures raises a problem that is often overlooked concerning logarithmically transformed data. A value reported as 10·0 kg, for example, implies that the range within which it is known is 0·10 kg, i.e., from 9·95–10·05 kg (Sokal & Rohlf, 1995). However, if 10·0 kg is converted to common logarithms and also reported with three significant figures as 1·00, this implies a range in logarithmic units of 0·995–1·005. The antilogs of these two values are 9·89 and 10·11, a range of 0·21 rather than 0·10. Precision has been "lost" by retaining a constant number of digits through the transformation and detransformation process. Thus, data converted to logarithms may need to be presented with more significant figures than would appear appropriate on the original scale. This need for additional significant figures also applies to the digits of the coefficients of regression equations calculated with logarithmically transformed data.

The definition of "adult"

The important issue in evaluating the meaning of "adult" for calculating adult body mass is identification of the cut-off point for inclusion of an individual in the data set. What separates adults from the previous stage, described either as juveniles, adolescents, subadults, or young adults? In addition, the question of whether or not there is a period of "old age" that should be excluded from an adult sample needs to be considered.

In studies of mammalian growth and development, a fully-adult animal is a normal individual (i.e., excluding those with pathologies of development) who has stopped growing. When human biologists speak of predicting adult height, it is the maximum height the individual will reach, not the height when third molars erupt or the height at which the individual has their first offspring (e.g., Tanner *et al.*, 1975). After a period of continuous, if irregular, increase, an individual primate reaches a body mass that remains more or less stable for an extended period of time, prior to a decline in old age (Kirkwood, 1985; Watts, 1986). It is the typical mass during this period of stability that we wish to capture when determining adult body mass. Some workers have attempted to identify this value with ontogenetic data by calculating the asymptote of a von Bertalanffy equation (Strum, 1991), or the inflection point of a lowess curve (Leigh, 1992*a*).

The lack of a clear distinction between the definition of adulthood and methods for identifying adulthood can lead to biases and errors in estimates of species' mean adult masses. For example, Altmann *et al.* (1981) defined the adult age class as "secondary sexual characteristics fully developed," but then, referring to Amboseli baboons, noted that "both males and females continue to grow for at least a year after entering full adulthood." The development of secondary sex characteristics, as well as the traits most commonly used to identify adults in skeletal collections, namely epiphyseal fusion of long bones or eruption of a complete permanent dentition, are strategies for identifying adults, not the definition of why that individual is an adult (Haigh & Scott, 1965; Michejda, 1978).

It appears to be common in primates for body mass to increase after third molar eruption (Phillips-Conroy & Jolly, 1981, 1988; Fooden & Izor, 1983), as well as after termination of limb growth (Sirianni & Swindler, 1985). Including all animals defined by these methods as adults in the calculation of adult body mass would result in underestimating adult mass and in different amounts of underestimation for different species. It is difficult to control this source of variation. However, when studies of a single species differed in the definition used to define adults (e.g., third molars emerged versus third molars at level of the occlusal plane), this information was considered in selecting data for inclusion in this study.

Although body mass decreases in old age, this has less of an impact on adult body mass values than the inclusion of younger animals that have not reached full size. First, old individuals are adults; the basis for excluding them is ambiguous. Second, their frequency in most samples is much less than that of young adults, and third, it appears that the loss of mass with old age is often relatively modest in magnitude (Turnquist & Kessler, 1989; Cheverud *et al.*, 1992). Nevertheless, in the few cases for which age-specific, cross-sectional data allowed identification of the age at which mean body mass decreased, those individuals were excluded from the means reported here.

Habitat

Body mass estimates in the literature exist for animals in laboratories (often with information about whether they were born in captivity or captured after birth), zoos (with or without "naturalistic" caging), semi-free ranging environments (e.g., Monkey Jungle, Duke University Primate Center), or natural habitats. There are further variations, including provisioned natural habitats (e.g., Sugiyama & Ohsawa, 1982), "food-enhanced" environments via access to human crops and/or refuse (e.g., Strum, 1991), and animals transported to a free-ranging environment that is not part of their natural range (e.g., Cayo Santiago).

There is evidence that some species show marked increases in body mass when kept in captive or provisioned habitats (Leigh, 1994). Captive orang-utans weigh on average 80% more than their wild-reared counterparts (Fooden & Izor, 1983), and obesity was common among captive gorillas before improved dietary management in recent years (Cousins, 1972; Leigh, 1992*b*). Altmann *et al.* (1993) found that free-ranging, 40+ month old female baboons that supplemented their diet with garbage were 52% heavier than wild-feeding females of the same age.

In this study, whenever possible, data from uncompromised natural habitats are reported. However, it should be noted that intraspecific population differences in the body mass of free-ranging animals can be affected by environmental variables such as mean annual temperature and rainfall (Popp, 1983; Dunbar, 1990; Patterson, 1996), and by differential productivity of ranging areas (Froehlich & Thorington, 1982). Many species also exhibit dramatic seasonal variation in body mass (DuMond & Hutchinson, 1967; Hladik *et al.*, 1980; Petter-Rousseaux, 1980; Zhao Qi-Kun, 1994; Wright & Martin, 1995).

Pregnancy and lactation

Reviews of body mass routinely attempt to exclude pregnant females. For some comparative questions, it would seem more appropriate to include them. Pregnancy and lactation place great energetic, physiological, and morphological demands on the body, and it is expected that the traits of females reflect adaptation to these demands. Nevertheless, including pregnant females raises many problems. These data are available for very few species. Females gain mass gradually throughout pregnancy, in many species adding more than 20% to total body mass by the end of gestation (Riopelle & Shell, 1978; Lunn, 1983; Deputte, 1991). If pregnant animals are included, should data be collected randomly for females, including whatever percentage happen to be in various reproductive states at a given time, or is the important value the maximum mass that females reach near the end of pregnancy? To provide a baseline against which the changes in mass during pregnancy can be measured, and to produce a data set of maximum cross-species comparability, pregnant females are excluded from this study whenever possible. Lactating females are excluded when reported masses differ from those of non-pregnant and non-lactating females.

Variability of body mass

In most statistical applications of body mass data, mean values are treated as absolute numbers measured without error. However, any estimate of body mass for a species in a sample estimate of a population mean. This estimate has uncertainty associated with it, usually best expressed as a standard error. The implications of variability in comparative statistical studies deserves much more attention than it has received.

Many studies providing data on species body mass include no information on variability, and many others follow the unfortunate tradition of reporting only the range as a measure of variation. When the standard deviation (or variance) of body mass is given, this statistic itself is a sample estimate of a population parameter. The standard error of measures of variation are relatively larger than the standard errors of measures of central tendency, so that if one wishes to estimate the standard deviation of body mass for a species, the small sample sizes common in the literature are even less adequate than they are for estimating means. There are insufficient data for us to present a useful table of primate body masses that includes standard deviations. Instead, we use the limited data from which reasonable estimates of variability can be obtained to make some broad generalizations concerning body mass variability. We selected for analysis only studies that included both sexes, and that had, with few exceptions, sample sizes larger than 30 for each sex. Data from laboratory animals are included in this analysis because of the advantages of large sample sizes. Table 1 shows the resulting database.

The coefficient of variation [CV = (Standard deviation/mean) × 100] for 19 species, calculated separately by sex, range from 4·3–19·5, with a mean of 12·5. There is no interpretable difference between the CVs of males and females (males 12·7, females 12·3). The overall mean CV coincides with Yablokov's (1974) suggestion that CVs for body mass of terrestrial mammals are typically 12–15. Because dimensionality affects the absolute magnitude of variation among organisms, the relative magnitude of CVs for linear, area, and volume measures in animals with similar shapes is approximately 1:2:3 (Yablokov, 1974; Lande, 1977), resulting in the seemingly large CVs for mass.

Although Yablokov (1974) found that there is often no relation between the mean and CV of body mass when comparing different species, a negative correlation between mass and the CV of mass can occur when measurement error is disproportionately large in smaller species (Lande, 1977), or in the unlikely event that the mean and standard deviation are themselves unrelated (Pearson, 1897). However, precisely the opposite is found here; a least squares equation for predicting CV from the natural log of body mass has a correlation of 0·570 ($P \approx 0{\cdot}02$ if $n = 19$, which would occur if species were independent and sexes within species added no degrees of freedom). The equation is:

$$\text{CV} = 1{\cdot}044\ (\ln \text{ body mass, kg}) + 11{\cdot}42$$

The standard error of estimate (SEE) is 2·69 and the standard error of the slope is 0·251. Although the prediction error is large, the general effect of size is substantial. For example, the equation predicts a CV of 9·7 for a species with a mean body mass of 200 g and a CV of 14·5 for a species with a mean body mass of 20 kg. These limited data suggest that there is a significant "positive allometry" (sensu Mosimann & James, 1979) between the CV and mass. We do not know if this finding is unique to primate body masses, and offer no reason here for why relative variation should necessarily increase with body size. However, we do note that sex dimorphism in body mass increases with increasing body mass in primates (Cheverud *et al.*,

Table 1 Data for analysis of intraspecific variation in primate body mass

	Males				Females				
	N	$\bar{X}$ (kg)	S.D.	C.V.	N	$\bar{X}$ (kg)	S.D.	C.V.	Reference
Galagoides zanzibaricus	35	0·150	0·0064	4·3	38	0·137	0·0061	4·4	Harcourt & Nash, 1986
Otolemur garnetti	120	0·794	0·0934	11·8	134	0·734	0·0818	11·2	Olson, 1979
Saguinus geoffroyi	55	0·482	0·0500	10·4	40	0·502	0·0550	10·9	Dawson & Dukelow, 1976
Saguinus oedipus	181	0·451	0·0621	13·8	171	0·473	0·0573	12·1	Cheverud *et al.*, 1994
Cebuella pygmaea	36	0·110	0·0110	10·0	27	0·122	0·0099	8·1	Soini, 1988
Alouatta caraya	58	6·42	1·15	17·9	117	4·33	0·764	17·6	Rumiz, 1990
Aotus seniculus	31	5·62	0·535	9·5	29	4·03	0·385	9·5	Braza *et al.*, 1983
Aotus nancymae	130	0·957	0·124	13·0	124	0·855	0·120	13·6	Baer, 1994
Saimiri sciureus	17	1·02	0·129	12·6	58	0·699	0·0778	11·1	Scollay, 1980
Cebus albifrons	24	3·26	0·429	13·2	43	2·13	0·282	13·2	Ausman *et al.*, 1982
Cercopithecus aethiops	26	4·13	0·56	13·6	36	2·57	0·25	9·7	Turner *et al.*, 1994
Cercopithecus mitis	41	5·85	0·918	15·7	94	3·93	0·622	15·8	Colyn, 1994
Macaca mulatta	35	11·9	1·7	14·3	40	9·6	1·3	13·5	Turnquist & Kessler, 1989
Papio anubis	39	21·2	2·1	9·9	35	11·7	1·3	11·1	Phillips-Conroy & Jolly, 1981
Papio hamadryas	41	16·9	1·9	11·2	39	9·9	1·3	13·1	Phillips-Conroy & Jolly, 1981
Hylobates lar	84	5·90	0·86	14·6	66	5·34	0·70	13·1	Geissmann, 1993
Hylobates syndactylus	89	12·8	2·5	19·5	87	10·5	1·7	16·2	Orgeldinger, 1994
Pan troglodytes	18	40·9	4·6	11·2	23	33·4	4·7	14·1	Uehara & Nishida, 1987
Homo sapiens	502	72·1	11·4	15·8	165	62·1	9·88	15·9	Holloway, 1980

1985; Leutenegger & Cheverud, 1982). This would explain some increase in variation for data sets in which sexes were pooled for calculating intraspecific variability. Our results are based on sexes treated separately, and therefore have no clear relationship to patterns of sex dimorphism.

For readers concerned that the relationship between CV and body mass involves regression of a ratio against its own denominator (Prothero, 1986), we calculated the relationship between the standard deviation and body mass for all 38 values in Table 1, with data transformed to natural logs to improve linearity and homoscedasticity. The least squares equation is:

$$\ln \text{ standard deviation} = 1{\cdot}102\ (\ln \text{ body mass}) - 2{\cdot}23$$

The correlation coefficient is 0·991, with a standard error of estimate of 0·259 and standard error of the slope of 0·024. The slope confirms the positive allometry of variation with increasing mass.[2]

Although data for calculating the variability of individual species is scarce, these results provide useful estimates of variability in primate body mass. For statistical procedures in which incorporating intraspecific variability would improve the validity of results, one could use the rough approximation that primate species have a CV for body mass of about 12·5. Alternatively, estimates based on body mass could be calculated from the above equations.

Evaluation of other resources

As Gittleman (1989) and others have pointed out, a comparative study can be no more reliable than the data on which it is based. During the literature search and cross-referencing of data for this study, shortcomings of some data sets became evident, some of which seriously compromise the validity of the published data, and others of which only qualify their use. Three sources are discussed here. Comments on the review of New World monkey masses by Ford & Davis (1992) have been made elsewhere (Smith, 1996).

Harvey & Clutton-Brock (1985)

The most widely-used, secondary source for primate body mass is the one published by Harvey & Clutton-Brock (1985) and the identical data set in Harvey *et al.* (1987). Among the major comparative studies that have used these body mass values are Kappler, 1990; Parker, 1990; Sawaguchi, 1990; Young *et al.*, 1990; Godfrey *et al.*, 1991; Ross, 1991; Hauser, 1993; Wrangham *et al.*, 1993; Aiello & Wood, 1994; Leigh, 1994; Harcourt *et al.*, 1995; and Janson & Goldsmith, 1995.

This set of body mass data is presented without references or sample sizes. We have always been surprised at the wide acceptance granted these values in the absence of documentation [e.g., Wrangham *et al.* (1993) described it as a "standard source"]. When the present project began, we hoped to use some of the data, and therefore, set out to understand their origins. The only information provided by Harvey *et al.* (1987) was (p. 182):

[2]If the values for *Galagoides zanzibaricus* are eliminated (a step for which we see no biological basis, but which was explored because these data had the largest statistical influence on the equation), the two equations are:

CV = 0·708 (ln body mass, kg) + 12·16 r = 0·44 SEE = 2·40

ln standard deviation = 1·055 (ln body mass) − 2·13 r = 0·995 SEE = 0·183

These changes do not alter the interpretations discussed above.

Table 2 **Values listed by Clutton-Brock & Harvey (1977) as "approximate figures taken in some cases from congeneric species of similar size belonging to the same ecological category" but listed without comment by Harvey & Clutton-Brock (1985)**

	Males	Females
Cebus nigrivittatus	—	2·30
Saimiri oerstedii	0·75	0·58
Ateles belzebuth	6·20	5·80
Ateles paniscus	—	5·80
Macaca sylvanus	—	10·00
Cercocebus torquatus	8·00	—
Papio cynocephalus	—	15·00
Papio papio	26·00	13·00
Cercopithecus campbelli	3·60	3·60
Presbytis geei	8·60	8·10
Presbytis rubricundus	6·30	6·30
Colobus angolensis	10·70	9·00

"The full data set, together with further details on data sources, definitions of variables, criteria for exclusion from our sample, and methods used for deriving species' values can be found elsewhere (Rudder, 1979; Harvey & Clutton-Brock, 1985)."

However, Harvey & Clutton-Brock (1985) provided no references for the body mass values, adding to our understanding only that (p. 567) "Data were taken from adults only. Care was taken to exclude sub-adult, pregnant, emaciated, or obese specimens," and Rudder (1979) stated only that (p. 19) "Most of the data used in analysis is already published and it was thought that such unoriginal material would be better presented elsewhere (Clutton-Brock, Harvey & Rudder, in prep., Rudder, in prep)."

Thus, the sources to which readers were directed provided no information on the origin of values for individual species. For some values, however, information is available by tracing sources not cited by Rudder (1979), Harvey & Clutton-Brock (1985), or Harvey *et al.* (1987).

In 1977, Clutton-Brock & Harvey published data on body masses for 91 primate species. In that paper, the values for both sexes of 32 species and for the females of three additional species were listed in brackets. In the explanation at the top of the table, Clutton-Brock & Harvey (1977) stated (pp. 34–35) "Where body weights are bracketed, they represent approximate figures taken in some cases from congeneric species of similar size belonging to the same ecological category." They went on to qualify *all* values in the table with the statement that (pp. 34–35) "The authors would like to stress that individual estimates shown in this Table may be unrealistic and that it should not be used as a reference source." Although the table in Harvey & Clutton-Brock (1985) contained no bracketed values, no admonition that it should not be used as a reference source, and no citation to Clutton-Brock & Harvey (1977), 82 of the 248 values (for 124 species), or 33% of the entire data set, were identical to those in the earlier table. Furthermore, 19 of these values were in brackets in 1977, but were simply listed as data in 1985 (Table 2). Thus, just under 8% of the values in the 1985 paper originated as "guesstimates" in 1977 but became facts 8 years later.

Problems are not limited to these 19 values which were originally "approximate figures." Examination of data in sources cited by Clutton-Brock & Harvey (1977) indicates that much more than 8% of the 1985 table represents values for which there has never been any data.

Table 3 **Body mass values for *Ateles* published by Clutton-Brock & Harvey (1977) and Harvey & Clutton-Brock (1985)**

	Clutton-Brock & Harvey, 1977		Harvey & Clutton-Brock, 1985	
	M	F	M	F
Ateles belzebuth	(6·2)	(5·8)	6·2	5·8
Ateles geoffroyi	6·2	5·8	6·2	5·8
Ateles paniscus	(6·2)	(5·8)	6·6	5·8
Ateles fusciceps	—	—	8·9	9·1

A. fusciceps was not listed in 1977 and is not part of this evaluation. Values in parentheses are described by Clutton-Brock & Harvey (1977) as "approximate figures taken in some cases from congeneric species of similar size belonging to the same ecological category." As discussed in the text, the values attributed to *A. geoffroyi* were not from individuals of this species.

Table 3 shows the values for three species of *Ateles* in Clutton-Brock & Harvey (1977). The use of brackets indicates that data were known for *Ateles geoffroyi*, but that both *Ateles belzebuth* and *Ateles paniscus* were simply estimated to be the same size as *A. geoffroyi*. Table 3 also shows how these data were listed in Harvey & Clutton-Brock (1985). *A. paniscus* males have a slightly different mass, but the other three values that were in brackets in 1977 were shown as facts. But where did the masses for *A. geoffroyi* listed in 1977 and repeated in 1985 come from in the first place? In Table 1 of Clutton-Brock & Harvey (1977), it is indicated that Napier & Napier (1967) "provided the majority of body weights." Also listed were all other sources of data for each species, so that if body masses were not included in these other references, the relationship to Napier & Napier is unambiguous. However, Napier & Napier generally provided sex-specific means and/or ranges of body masses for genera, with no breakdown of data by species. For the entire genus *Ateles*, Napier & Napier reported data for three individuals, two males of 5470 g and 6887 g (an average of 6178 g), and one female of 5824 g. Given the other sources of information for *Ateles* in Clutton-Brock & Harvey (1977), there is no alternative to the conclusion that these were rounded to 6·2 and 5·8 as the source for *A. geoffroyi*. Because it is not evident from Napier & Napier which species of *Ateles* these masses referred to, we checked the references cited by Napier & Napier. All three values were from Hill (1962). The two males were *A. paniscus* (p. 498) and the one female was an *A. belzebuth* (p. 494). Thus, there were no measurements of *A. geoffroyi* in the data that Clutton-Brock & Harvey (1977) attributed to that species. Excluding the minor change in *A. paniscus* males (which itself is not close to any values we have found for *A. paniscus* males), the six species "means" in Harvey & Clutton-Brock (1985) were based on a total of three animals.

This misuse of generic data from Napier & Napier (1967) for *Ateles* is not an isolated case. Napier & Napier gave values of 7392 g (n=4) and 5720 g (n=4) for males and females, respectively, of the genus *Alouatta*, for which they listed five species. Clutton-Brock & Harvey (1977) and Harvey & Clutton-Brock (1985) showed these as masses of 7·4 and 5·7 for *Alouatta palliata*. But reviewing Napier & Napier's sources, all eight values were for *Alouatta seniculus*, from Sanderson (1949). In this case real data for *A. palliata* were available before 1985 in Scott *et al.* (1976), Milton *et al.* (1979), Thorington *et al.* (1979), and Estrada (1982), none of which Harvey & Clutton-Brock (1985) used.

Thus, the Harvey & Clutton-Brock (1985) data [as well as the Harvey *et al.* (1987) data] have three major problems: (1) estimated values from 1977 were treated as real data, (2) generic

data from Napier & Napier (1967) were attributed incorrectly to species, and (3) the literature review was inadequate, so that many values unaffected by the first two problems did not represent the best of what was known at the time. In addition to these issues, there remain oddities in the Harvey & Clutton-Brock (1985) data set, by which we mean body masses that differ in their values or relationships from any data that we can find in the literature and that are not explained by any of the above three problems. For example, sex dimorphism of 15% or more were reported for *Lemur catta*, *Lemur fulvus*, and *Varecia variegata*, and no dimorphism at all for *Cebus albifrons* and *Callicebus torquatus*. Reasonably good data for these species indicate exactly the opposite—negligible or reverse sex dimorphism in the three species of Lemuridae, and sex dimorphism greater than 15% in both *Cebus* and *Callicebus*. Harvey & Clutton-Brock (1985) also indicated that *Papio ursinus* males were nearly the smallest of all *Papio* males, when they are distinctly the largest. The values given for *Homo sapiens*, 47·9 and 40·1 kg, are ideal for an African pygmy population.

For all of these reasons, we conclude that the body masses provided by Clutton-Brock & Harvey (1977), Harvey & Clutton-Brock (1985), and Harvey *et al.* (1987) should not be used.

Leutenegger & Cheverud (1982)

Leutenegger & Cheverud (1982) published data for 53 species, which has occasionally been a source for other workers. The legend for this table suggests that it is an excellent sample (p. 391):

> "Each of the 53 species is represented by male and female means of a total of at least 16 individuals, with approximately equal numbers of each sex. Although the data are taken from various sources, and are therefore not perfectly comparable, a major source of error usually found in body weight studies has been eliminated by disregarding data from captive animals and using data only from wild-trapped and wild-shot animals. All the body weight data were extracted from Leutenegger (1973, 1978), Rothenfluh (1976), and Leutenegger & Kelly (1977)."

However, every one of the 106 values listed by Leutenegger & Cheverud (1982) is identical to the value listed for free-ranging specimens in Rothenfluh (1976). (There are three typographical errors. Rothenfluh lists *Alouatta caraya* females at 4983 g, *Hylobates agilis* females at 5550 g, and *Perodicticus potto* males at 1225 g. Leutenegger & Cheverud (1982) listed these as 4893, 5500, and 1125, respectively.) Furthermore, in Rothenfluh's tables, 17 of the values listed by Leutenegger & Cheverud are based on sample sizes of one or two animals, and four on an unknown numbers of animals.

The data collected by Rothenfluh (1976) appear valid and useful, so that the limitations of Leutenegger & Cheverud (1982) as a source for body masses are of an entirely different, and less significant, nature than those of Harvey & Clutton-Brock (1985). Simply, these body mass data are more dated and have lower reliability (because of smaller sample sizes) than indicated.

Silva & Downing (1995)

Recently, Silva & Downing (1995) published data on body mass for all mammalian orders, listing over 1600 references. Included are 19 pages of data on primates, covering what the authors identify as 69% of known primate species. Although not yet cited in the primatological literature, this appears at first glance to be a new and promising resource.

Silva & Downing listed data separately by location when multiple sites for a species have been reported in the literature. Unfortunately, the uncritical use of this approach has resulted in the listing of data that never existed. For example, Freese *et al.* (1982) conducted a census of primates at 14 sites in Peru and Bolivia. In order to estimate species biomass at each site, Freese *et al.* (1982) used approximate adult masses for 15 species. They took these values

primarily from Napier & Napier (1967), and clearly used them as rough approximations, so that *Saguinus fuscicollis*, *Saguinus imperator*, *Saguinus labiatus*, *Saguinus nigrocollis* and *Callithrix argentata* were all listed as weighing 0·35 kg. However, Silva & Downing used Freese *et al.* not only as a source of data on body mass for all of these species, but as a source of data on geographic variation in body mass. Thus, they reported a body mass of 0·35 kg for *S. fuscicollis* 11 separate times, at the 11 different locations at which Freese *et al.* reported the presence of the species. Of course, no data on mass were collected for any specimens at any of these sites.

Elsewhere, Silva & Downing listed a value of 278 kg for male gorillas, as well as an average for male and female gorillas of 80 kg. *Papio anubis* was shown in two entries to have a mean sex combined body mass of 33·3 kg and of 7·9 kg. *Galago senegalensis* had listings of 1·56 and 1·51 kg for males and females, respectively, and of 0·156 and 0·151 kg. Moreover, these two entries for *G. senegalensis* were duplications of a single data set, appearing twice in the literature because the authors cited their own previous work (Smithers, 1971; Skinner & Smithers, 1990). Finally, many key references were omitted. The only data for orang-utans, for example, were from Bauchot & Stephan (1969). In sum, while Silva & Downing (1995) may contain some useful bibliographic citations, the data themselves will probably find no use among primatologists.

Differences among data sets

The consequences of using different sets of body mass data on the results of comparative analyses will vary with the particular species used and with the statistical procedures employed. Table 4 is a modified stem-and-leaf display (Tukey, 1977) in which each value represents a ratio calculated as the body mass from the Harvey & Clutton-Brock (1985) data set divided by the comparable value from our Table 5. Values were selected from Table 5 in order to minimize differences between the two data sets. For example, Harvey & Clutton-Brock (1985) reported masses for *Nycticebus coucang* of 1·30 and 1·20 kg. We, therefore, used masses only for *N. coucang bengalensis* in Table 5 to calculate ratios. We also used 0·062 and 0·063 kg as their values for *Galagao demidovii*, rather than the listed typographical errors of 0·62 and 0·63 kg. As shown, the ratios for 113 species range from 0·68–1·81 for males and from 0·55–1·76 for females. Differences of these magnitudes clearly can alter the interpretation of results from comparative studies. Similar ratios calculated with the Leutenegger & Cheverud (1982) data and Table 5 range from 0·66–1·35 for males and from 0·67–1·44 for females.

There are other types of analyses that either magnify or minimize the consequences of differences among data sets. Perhaps the smallest effect on a comparative study occurs when the very large range of primate body masses comes into play and covariation is the statistical feature of interest. When the full range of body masses from *Microcebus* to *Gorilla* are included, the Pearson product–moment correlation coefficient between the values reported by Harvey & Clutton-Brock (1985) and our Table 5 is 0·994 and 0·993 for males and females, respectively, of 113 species (data converted to logarithms). It has been recognized for some time that even correlations of this magnitude allow for wide variation of individual values around a regression line when data are distributed over a large range (Smith, 1980).

Materials and methods

Several issues in data collection are addressed in preceding sections. Discussion of our procedures regarding significant figures, defining the adult age category, the habitats of animals, and the treatment of data on pregnant females, are included in the sections dealing with each of these topics.

Table 4 Stem-and-leaf display of differences in body mass data from Harvey & Clutton-Brock (1985) and this study

Leaves (Females)	Stem	Leaves (Males)
5	0·5	
8	0·6	8
4 3	0·7	1 2 2 4
9 8 6 6	0·7	5 8 9 9 9
4 3 2 2	0·8	0 0 0 1 1 1 2 3 3 3
9 9 8 8 8 8 7 5 5 5	0·8	5 5 7 7 8 8
4 4 2 2 2 2 1 1 1 0	0·9	0 1 1 1 2 2 3 3 4 4
9 9 9 9 9 8 8 8 7 7 6 6 6 5 5 5	0·9	5 5 5 6 6 7 7 8 8 8 8 8 9 9
4 3 3 3 3 3 2 2 2 2 2 1 1 1 0 0 0 0 0 0 0 0 0	1·0	0 0 0 0 0 0 0 0 0 0 1 1 1 2 2 2 2 3 3 3 3 3 4 4 4
9 8 7 7 6 6 6 6 6 5 5	1·0	5 5 5 5 6 6 6 6 7 7 7 7 9 9
4 4 3 3 3 3 3 3 2	1·1	0 0 0 1 2 3 3 4 4
9 9 8 8 7 6 6 5	1·1	6 9
1 0	1·2	2 2 3
9 6 6	1·2	6 6
4 3 2 1 0	1·3	0 0 1
6 5	1·3	5
0	1·4	1 2 5
3	1·5	
	1·6	
6	1·7	
	1·8	0 1

Each value is a ratio of the body mass from Harvey & Clutton-Brock (1985) divided by the body mass from Table 5 of this study. The stem indicates the first digit(s) of the ratio, and each value in the Leaves columns identifies the last digit of a ratio. For example, the highest ratio for females is 1·76, and the second highest is 1·53. With males, there is one species with a ratio of 1·80 and another with a ratio of 1·81.

Taxonomy and data collection

One of the most difficult issues in compiling a data set of body masses for primates is the complication created by changing taxonomy. Consider, for example, the genus *Aotus*. Until fairly recently, this genus was considered to have only one species, *Aotus trivirgatus*, with geographical variation assigned to subspecies. Today, while the taxonomy of this group remains actively discussed (Ford, 1994), 10 species of *Aotus* are recognized for this study. A reference in the literature to body mass for "*A. trivirgatus*" cannot be assumed to apply to the species considered to be *A. trivirgatus* today. Unless older studies indicate sub-species, these potentially valuable data cannot be used. Another example is the genus *Arctocebus*, which contains two species, *Arctocebus aureus* and *Arctocebus calabarensis*. Much of the data on *A. aureus* was collected at a time when *A. aureus* was considered to be *A. calabarensis aureus*, and body mass values for *A. aureus* are sometimes listed in publication without indication of subspecies, therefore appearing simply as *A. calabarensis* (e.g., Bearder, 1987).

For this study, we began by first creating a list of all extant primate species. In general, when in doubt, splitting was prioritized over lumping. This does not reflect taxonomic decisions or philosophy. Given mean values and sample sizes, split data can be easily lumped, while lumped data cannot be split. To maximize the flexible use of the data, most species were divided when there was any question in the recent literature. We then created a data file for each species, in which we listed every body mass value found. Discrepant data within species and overlapping data among congeneric species provided clues to many of the taxonomic

errors that might occur. Although much effort was expended to minimize this problem in the data presented here, it is likely that some incorrect assignments remain. It should also be noted that all citations are to articles that have been seen directly by the authors. There are no citations of citations from secondary sources. Secondary sources are used and cited when the original literature could not be obtained, or when the secondary source is an analytical review of data. For a very few species, the only data of any kind on body mass are in secondary (or tertiary) sources with no citations or sample sizes (e.g., Haltenorth & Diller, 1980). Our data for *H. sapiens* are but a few samples from a vast literature (e.g., Eveleth & Tanner, 1990).

The species list is alphabetical within families, except for the separation of Cercopithecinae from Colobinae, and is from Groves (1993), with the following modifications. (1) Lemuroidea follow Mittermeier *et al.* (1994). (2) Galagos are treated as a subfamily (Rasmussen & Biela, 1997) and listed in Lorisidae, with the addition of *Galagoides thomasi* (Nash *et al.*, 1989) and *Pseudopotto martini* (Schwartz, 1996). (3) Within the New World monkeys, the genus *Cebuella* is retained for *C. pygmaea*, and several new species are recognized, including *Callithrix emiliae* (Mittermeier *et al.*, 1988), *Callithrix mausi* (Mittermeier *et al.*, 1992), *Callithrix nigriceps* (Ferrari & Lopes, 1992), *Callithrix saterei* (Silva & Noronha, 1996), and *Cebus kaapori* (Ferrari & Queiroz, 1994). (4) Within Cercopithecinae, Fooden (1976*a*) is followed for genus *Macaca*, the genus *Cercopithecus* (rather than *Chlorocebus*) is used for *C. aethiops*, *Papio* is treated as five species rather than as one (Napier, 1981), and both *Cercocebus atys* and *Lophocebus aterrimus* are recognized as species. (5) Finally, Colobinae follow Oates *et al.* (1994), with the addition of *Presbytis femoralis* (Aimi & Bakar, 1992).

Pooling data

All data in Table 5 are for free-ranging animals except where specifically indicated in the "Comments" column of the table. However, body masses of free-ranging animals reported in primary sources may either be taken from live animals, in which case individuals often are from a single locale and may represent an interbreeding population, or they may be taken from museum collections, in which case the specimens were usually collected over a range of time and locations. Thus, even data limited to free-ranging animals can vary in the extent to which geographic differentiation is represented.

Sample means for a species are also affected by the criteria and procedures used to combine data from multiple studies. There may seem to be appeal in the idea of collecting all possible data and finding a grand, overall mean for each sex of a species. However, this approach can lead to serious errors, particularly when estimating sex dimorphism within species. It is well documented that body size can vary considerably between habitats for a single species. For example, it appears that *A. palliata* on Barro Colorado Island (BCI) are larger than those in Costa Rica (Scott *et al.*, 1976; Milton *et al.*, 1979; Glander *et al.*, 1991). Although within each location, males are substantially larger than females, the geographic differences are so great that the females on BCl may be as large as the males in Costa Rica. If data sets from each location are combined to determine an overall mean, but the number of males and females *within* each data set is not identical, it is possible to end up with sex-specific means that indicate an overall sex dimorphism markedly different from the dimorphism within either population. For this reason, we reject the strategy of combining all available values to determine a species mean. An additional reason not to combine all available data is that often there are substantial differences in quality among studies. Finally, museum studies are not combined if it appears likely that the investigators used the same museums. Care must also be taken in combining masses reported in the literature from field expeditions with measurements from museum

specimens. The animals shot and weighed during these expeditions are often the same specimens evaluated decades later in museums.

We emphasize that the data presented in Table 5 are not meant to be a specialist-level review of sub-specific and geographic variation in body mass. Our preference is to list one pair of values for each species, for use in comparative studies by researchers who have been using data from sources such as Jungers (1985), Harvey & Clutton-Brock (1985), or Fleagle (1988). When possible, the means reported in this study are for single, large populations with well documented methods. Our strategy is to use the best available data rather than all available data. When different studies indicate possible important geographic differences in body mass within species, data are listed separately rather than combined. Data are combined when differences between populations or samples appear to be explained by the variability that is expected within populations (i.e., as sampling differences), or when none of the data represents an identifiable population (such as data from several museum studies). Also, in some cases, data are known for a small sample of free-ranging animals and for a substantially larger sample from a less desirable habitat. Although it is possible to choose between these samples and report only one, it can provide an important perspective to know that a large laboratory sample confirms (or contradicts) the values from a small field study. The decision to present multiple listings for a species is subjective and reflects our personal judgement, as does the selection of data when many good studies are available for a single species. The following examples clarify how some of these choices were made.

Callithrix jacchus. Data are listed for a large captive colony and for a small sample of free-ranging individuals. This is not because the free-ranging animals are about 15% larger. With an expected CV of about 12, data sets that differ by 15% are within one standard deviation of each other, and are therefore reasonable sampling differences. However, if only the data from free-ranging individuals were listed, the small sample size would lead to low confidence in the values, and if only the captive data were listed, the effects of captivity would remain a question. The same considerations were involved in selecting the data listed for *Cheirogaleus medius, Eulemur coronatus*, and *Aotus lemurinus*, among others.

Chiropotes satanus. Here the difference between the two subspecies is not large, and would in most cases lead us to combine data as a single entry. However, female *C. s. satanus* are heavier than male *C. s. chiropotes*, even though within each subspecies, males are larger than females. This may be important in studies concerned with sex dimorphism. Many other species exhibit similar patterns, including *A. palliata, A. seniculus, Lagothrix lagotricha, Cercopithecus aethiops, Procolobus badius*, and *Pan troglodytes*.

Papio anubis. For this "species" (Jolly, 1993), the issue involves selecting data from among many studies. Simplifying discussion by looking only at males, good data (sample sizes of 10–50 for each sex, free-ranging animals, adult status defined), range from 27·1 (Popp, 1983) to 21·2 kg (Phillips-Conroy & Jolly, 1981), in mean body mass. An overall average in the area of 24 kg would not be misleading. However, the low value of Phillips-Conroy & Jolly (1981) is from a population that has been studied extensively. It would not be surprising to find data on home range, diet, or other features of these Awash baboons used to represent *P. anubis* in a comparative study. In order to reduce the possibility of matching these values with the significantly higher species mean body mass, we list data for the Awash population. The value of 27·1 kg, however, is higher than several other good data sets from similar areas, and because

our objective is to represent the species, not to indicate the extremes of the range, we pooled data from populations in which males had values of 27·1, 22·8 (Popp, 1983), 23·8 (Eley *et al.*, 1989), and 21·9 kg (Gest & Siegel, 1983). The Awash *P. anubis* were excluded from calculation of a species mean because they are on the periphery of the geographic range of olive baboons, and their size probably reflects their relatively impoverished environment.

Every listing in Table 5 was made after similar explicit evaluations of what would be most useful and representative given the available data for a species, keeping in mind the objective of publishing a broad-scale rather than specialized listing.

Results

For reasons outlined in previous sections, particularly (1) variation in body mass among populations within species, (2) variation in quality among studies reporting body mass, and (3) variation in sample sizes, potentially biasing the estimation of sex dimorphism, we emphasize that the best estimates of species body mass for broad general use in comparative studies are not determined by simply combining all available data for a species. Although we attempted to comprehensively review the literature for this study, the data in Table 5 are a subset of those data, chosen on the basis of criteria outlined in preceding sections. Thus, Table 5 itself is not a comprehensive listing of available literature.

A data entry with no comments should be understood to represent a general species-wide estimate based on free-ranging animals. Additional entries indicate special populations, significant contradictory or confirmatory data, sub-specific differences, or other information that may be important. The entire species list is presented in Table 5, so that readers will be aware of species for which no body mass data are known. Except where indicated, all reported values are arithmetic means. When studies are combined, the individual means are weighted by sample size for the calculation of the species values. Additional information on the masses identified in Table 5 as collected for "this study" are presented in the appendix.

Discussion

Comparative quantitative studies concerned with patterns of interspecific variation often involve bringing together three distinct types of data. First, obviously, is the primary topic of interest, such as home range (Milton & May, 1976), brain size (Armstrong, 1985), metabolic rate (Müller, 1985; Ross, 1992), group size (Wrangham *et al.*, 1993), litter size (Chapman *et al.*, 1990), limb proportions (Jungers, 1985) or tooth size (Cochard, 1987; Greenfield & Washburn, 1991), among many. These data typically represent an area of special interest for the investigator. If obtained from the published literature, it is common to see a careful and critical eye applied to the problem of identifying reliable and valid values.

A second data set needed in many studies is information on the phylogenetic relationships among the species used. This information is necessary in order to use statistical methods that take into account the non-independence of species (Felsenstein, 1985; Martins, 1996). The laboratory methods of molecular biology used to generate much of the best phylogenetic information are unfamiliar to most comparative primatologists. Therefore, to a great extent, these data are used with limited critical evaluation. Nevertheless, it appears that usually they are selected with care. The study of phylogenetic relationships is a vigorous and fast-moving topic. This is generally recognized by those using this literature and the need to be aware of the most current sources is widely appreciated.

Table 5 Body mass of extant primates

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Cheirogaleidae						
Allocebus trichotis	2	2	0·092	0·084	Captive	Meier & Albignac, 1991
Cheirogaleus major	3	6	0·438	0·362		Stephan & Bauchot, 1965; Wright & Martin, 1995; L. Martin, pc
Cheirogaleus medius	23	25	0·283	0·282	Duke Primate Center	Kappeler, 1991
	7	6	0·188	0·172	avg. over 2 seasons, interpolated from graph	Hladik *et al.*, 1980
Microcebus murinus	37	126	0·059	0·063	Breeding season,	Jolly, 1984
	9	11	0·125	0·111	non-breeding season, captive	Glatston, 1979
Microcebus myoxinus	15	17	0·031	0·030	Interpolated from graph	Schmid & Kappeler, 1994
Microcebus rufus	55	32	0·043	0·042	Average over two seasons	Atsalis, 1997
	7	8	0·082	0·092	Non-breeding season, captive	Glatston, 1979
Mirza coquereli	10	13	0·304	0·326	Average over two seasons	Stanger *et al.*, 1995
Phaner furcifer	3		0·460		Captive	Petter-Rousseaux, 1980
Megaladapidae						
Lepilemur dorsalis			≈0·500			Harcourt & Thornback, 1990
Lepilemur edwardsi	1	4	0·908	0·934		R. Warren, pc; this study
Lepilemur leucopus	2	5	0·617	0·594		Russell, 1977
Lepilemur microdon	1		0·970			L. Porter, pc
Lepilemur mustelinus	?		0·777			Eaglen, 1986
Lepilemur ruficaudatus	10	10	0·761	0·779		Bauchot & Stephan, 1966; Schmid & Ganzhorn, 1996; J. Ganzhorn, pc
Lepilemur septentrionalis			≈0·750			Harcourt & Thornback, 1990
Lemuridae						
Eulemur coronatus	2	2	1·28	1·08		Terranova & Coffman 1997; Terranova, pc
	16	14	1·69	1·62	Duke Primate Center	
Eulemur fulvus	6	5	1·95	2·15	*E. f. albocollaris*	Bradley *et al.*, 1997
	20	13	2·18	2·25	*E. f. rufus*	Glander *et al.*, 1992
	2	2	1·87	1·83	*E. f. sanfordi*	Terranova & Coffman, 1997; Terranova, pc
Eulemur macaco	2	5	1·88	1·76	*E. m. flavifrons*	Terranova & Coffman, 1997; Terranova, pc
	46	41	2·37	2·51	Duke Primate Center	
Eulemur mongoz	4	4	1·41	1·56		Terranova & Coffman, 1997; Terranova, pc
	34	33	1·63	1·61	Duke Primate Center	
Eulemur rubriventer	12	17	1·98	1·94		Glander *et al.*, 1992; Terranova & Coffman, 1997; Terranova, pc
Hapalemur aureus	4	2	1·52	1·39		Meier *et al.*, 1987; Glander *et al.*, 1992

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Hapalemur griseus	2	2	0·748	0·670	*H. g. griseus*	Terranova & Coffman, 1997; Terranova, pc
	11	14	0·987	0·903	*H. g. griseus*, Duke Primate Center	Kappeler, 1991
	4		1·45		*H. g. alaotrensis*	Mutschler & Feistner, 1995
Hapalemur simus	2	1	2·15	1·30		Kappeler, 1991
Lemur catta	41	24	2·21	2·21		Sussman, 1991
Varecia variegata	13	5	3·63	3·52	*V. v. variegata*	Terranova & Coffman, 1997; Terranova, pc
	46	35	3·47	3·51	Duke Primate Center	Kappeler, 1991
Indriidae						
Avahi laniger	4	4	1·03	1·32		Glander *et al.*, 1992
Avahi occidentalis	6	3	0·814	0·777		Bauchot & Stephan, 1966; R. Warren, pc
Indri indri	2	3	5·83	6·84		Bauchot & Stephan, 1966; Powzyk, 1996
Propithecus diadema	13	14	5·94	6·26		Glander *et al.*, 1992; Powzyk, 1996; this study
Propithecus tattersalli	10	8	3·39	3·59		Ravosa *et al.*, 1993
Propithecus verreauxi	3	4	3·25	2·95	*P. v. verreauxi*	Ravosa *et al.*, 1993
	10	5	3·70	4·28	*P. v. coquereli*	
Daubentoniidae						
Daubentonia madagascariensis	13	14	2·62	2·49		Glander, 1994; Feistner & Sterling, 1995
Lorisidae						
Arctocebus aureus	30		0·210			Charles-Dominique, 1977*b*
Arctocebus calabarensis	3	9	0·312	0·306		Jewell & Oates, 1969; this study
Euoticus elegantulus	5	3	0·287	0·261		Jewell & Oates, 1969; Chivers & Hladik, 1980; T. Olson, pc
	39		0·300			Charles-Dominique, 1974
Euoticus pallidus						
Galago alleni	9	30	0·277	0·269		Charles-Dominique, 1977*a*
Galago gallarum	1		0·200			J. Masters, pc
Galago matschiei	2	4	0·207	0·212		T. Olson, pc
Galago moholi	112	81	0·187	0·173		Skinner & Smithers, 1990; Harcourt & Bearder, 1989
Galago senegalensis	8	9	0·315	0·250	*G. s. braccatus*	Cooper, 1966 cited by Izard & Nash, 1988
	80	67	0·227	0·199		T. Olson, pc
Galagoides demidoff	54	19	0·063	0·060		Charles-Dominique, 1972
Galagoides thomasi	2	1	0·103	0·130		T. Olson, pc
Galagoides zanzibaricus	34	38	0·149	0·137		Harcourt & Bearder, 1989

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Loris tardigradus	7	4	0·264	0·269	*L. t. lydekkerianus*	P. Jenkins, pc
	10	8	0·192	0·193	*L. t. malabaricus*, Duke Primate Center	Kappeler, 1991
Nycticebus coucang	56	44	0·679	0·626	*N. c. coucang*	Barrett, 1984; Bearder, 1987
	4	2	1·10	1·02	*N. c. bengalensis*	This study; Fooden, 1971
Nycticebus pygmaeus	9		0·307			Barrett, 1984; Tan, 1994
	7	5	0·462	0·376	Duke Primate Center	Kappeler, 1991
Otolemur crassicaudatus	66	35	1·19	1·11		Skinner & Smithers, 1990; T. Olson, pc
Otolemur garnettii	120	134	0·794	0·734	Zanzibar & Pemba Island	Olson, 1979
Pseudopotto martini						
Perodicticus potto	5	5	1·25	1·21	*P. p. edwardsi*	Malbrant & Maclatchy, 1949; Stephan & Bauchot, 1965; this study
	33		1·10		*P. p. edwardsi*	Charles-Dominique, 1977*b*, 1978
	17	15	0·830	0·836	*P. p. potto*	This study
Tarsiidae						
Tarsius bancanus	21	16	0·128	0·117		Niemitz, 1984
Tarsius dianae	1	2	0·104	0·107		Niemitz *et al.*, 1991; Tremble *et al.*, 1993
Tarsius pumilus	Head and body length ≈75% of *T. spectrum* and *T. syrichta*					Musser & Dagosto, 1987
Tarsius spectrum	14	25	0·125	0·108		Nietsch & Niemitz, 1992; S. Gursky, pc
Tarsius syrichta	10	17	0·134	0·117	Duke Primate Center	Kappeler, 1991
Callitrichidae						
Callimico goeldii	6	7	0·499	0·468	Captive	Rothenfluh, 1976
	6		0·582		Jersey Wildlife Preserv. Trust	Carroll, 1982
Callithrix argentata	8	10	0·330	0·360		Ayres, 1986
Callithrix aurita	3		0·429			Muskin, 1984
Callithrix emiliae	12	5	0·313	0·330	Mid-range for females	Ferrari & Lopes, 1992; Ford & Davis, 1992
Callithrix flaviceps	4		0·406			Rosenberger, 1992
Callithrix geoffroyi	46		0·359			Rosenberger, 1992
Callithrix humeralifer	15	13	0·475	0·472	After air shipping	Kingston & Muniz, 1983
	4	5	0·36	0·38		Ayres, 1986
Callithrix jacchus	>30	>50	0·317	0·324	Captive	Turton *et al.*, 1978
	3	2	0·362	0·381		Epple, 1970
Callithrix kuhlii	≈0·375					Rylands, 1989
Callithrix mauesi	2	2	0·345	0·398		Mittermeier *et al.*, 1992
Callithrix nigriceps	3	1	0·370	0·390		Ferrari *et al.*, 1993
Callithrix penicillata	8	8	0·344	0·307		This study

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Callithrix saterei						
Cebuella pygmaea	36	27	0·110	0·122		Soini, 1988
Leontopithecus caissara		1		0·572		Ford & Davis, 1992
Leontopithecus chrysomelas	2	6	0·620	0·535		Rosenberger & Coimbra-Filho, 1984
Leontopithecus chrysopygus	4		0·575			Rosenberger & Coimbra-Filho, 1984
Leontopithecus rosalia	>50?	>30?	0·620	0·598	Reweighed seasonally	Dietz *et al.*, 1994
Saguinus bicolor	4	4	0·428	0·430		Ayres, 1986; Ferrari, 1993
Saguinus fuscicollis	69	55	0·343	0·358		Snowdon & Soini, 1988
Saguinus geoffroyi	55	40	0·482	0·502		Dawson & Dukelow, 1976
Saguinus imperator	4	1	0·474	0·475		Peres, 1993; Ferrari, 1993
Saguinus inustus	2	1	0·585	0·803(?)		Hernandez-Camacho & Defler, 1985; Ferrari, 1993
Saguinus labiatus	136	77	0·490	0·529		Puertas *et al.*, 1995
Saguinus leucopus	2	2	0·494	0·490		Hernandez-Camacho & Defler, 1985
Saguinus midas	34	15	0·515	0·575		Fleagle & Mittermeier, 1980; Ayres, 1986; Ferrari, 1993; this study
Saguinus mystax	191	133	0·510	0·539		Snowdon & Soini, 1988; Garber *et al.*, 1993
Saguinus nigricollis	8	6	0·468	0·484		Hershkovitz, 1977; Hernandez-Camacho & Defler, 1985
Saguinus oedipus	37	29	0·418	0·404	*S. o. oedipus*	Savage *et al.*, 1993
Saguinus tripartitus						
Cebidae						
Alouatta belzebul	27	26	7·27	5·52		Peres, 1994
Alouatta caraya	58	117	6·42	4·33		Rumiz, 1990
Alouatta coibensis						
Alouatta fusca	4	5	6·73	4·35		This study
Alouatta palliata	110+	177+	7·15	5·35		Peres, 1994
	14	18	6·53	4·02	Costa Rica	Glander *et al.*, 1991
	15	15	7·8	6·6	Panama	Thorington *et al.*, 1979
Alouatta pigra (=villosa)	2	4	11·4	6·43		Peres, 1994
Alouatta sara						
Alouatta seniculus	81	76	6·69	5·21		Ayres, 1986; Rudran, 1979; Braza *et al.*, 1983; Hernandez-Camacho & Defler, 1985
	64	46	6·31	4·67	Venezuela	Rodriguez & Boher, 1988
	28	34	7·62	6·02	Brazil	Ayres, 1986

Table 5 ***Continued from previous page***

	Sample size		Body mass, kg			
	M	F	M	F	Comments	References
Aotus azarae	4	8	1·18	1·23		This study
Aotus brumbacki						
Aotus hershkovitzi						
Aotus infulatus	1	1	1·19	1·24		Fernandes, 1993
Aotus lemurinus	18	12	0·918	0·874		Crile & Quiring, 1940; Hernandez-Camacho & Defler, 1985
	127	101	0·908	0·889	Laboratory colony	Dixson, 1983; Baer, 1994
Aotus miconax						
Aotus nancymae	32	24	0·794	0·780		Aquino & Encarnacion, 1986
Aotus nigriceps	1	2	0·875	1·04		Peres, 1993
Aotus trivirgatus	20	17	0·813	0·736		This study
Aotus vociferans	20	20	0·708	0·698		Montoya *et al.*, 1990
Ateles belzebuth	12	20	8·29	7·85		Ayres, 1986; this study
Ateles chamek	4	8	9·41	9·33		J. Kelley, pc
Ateles fusciceps	6	11	8·89	9·16		Crile & Quiring, 1940
Ateles geoffroyi	25	63	7·78	7·29		Schultz, 1941; Glander *et al.*, 1991; T. Cole, pc
Ateles marginatus						
Ateles paniscus	20	42	9·11	8·44		Smith, 1996
Brachyteles arachnoides	4	3	9·61	8·07		Lemos de Sa & Glander, 1993
Cacajao calvus	1	2	3·45	2·88		Ayres, 1986
Cacajo melanocephalus	5	6	3·16	2·71		Smithsonian Computer Database; Ayres, 1986
Callicebus brunneus	6	4	0·854	0·805	Mid-range	Ford & Davis, 1992
Callicebus caligatus	1		0·880			Ferrari & Lopes, 1995
Callicebus cinerascens						
Callicebus cupreus	10	2	1·02	1·12		Hershkovitz, 1990; Peres, 1993
Callicebus donacophilus	5	2	0·991	0·909		This study
Callicebus dubius						
Callicebus hoffmannsi	1	4	1·09	1·03		Ford & Davis, 1992; this study
Callicebus modestus						
Callicebus moloch	10	19	1·02	0·956		Ayres, 1986; Hershkovitz, 1990; Ferrari & Lopes, 1995
Callicebus oenathe						
Callicebus olallae						
Callicebus personatus	5	7	1·27	1·38		Hershkovitz, 1990
Callicebus torquatus	15	21	1·28	1·21		Hernandez-Camacho & Defler, 1985; Hershkovitz, 1990; this study

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Cebus albifrons	26	15	3·18	2·29		Hernandez-Camacho & Defler, 1985; this study
Cebus apella	51	38	3·65	2·52		Ayres, 1986; Fleagle & Mittermeier, 1980; Hernandez-Camacho & Defler, 1985; this study
Cebus capucinus	16	10	3·68	2·54		Crile & Quiring, 1940; Schultz, 1941; Glander *et al.*, 1991; Hernandez-Camacho & Defler, 1985
Cebus kaapori						
Cebus olivaceus (=nigrivittatus)	28	10	3·29	2·52		Rodriguez & Boher, 1988; this study
Chiropotes albinasus	7	7	3·15	2·49		Hershkovitz, 1985
Chiropotes satanas	20	19	2·90	2·58	*C. s. chiropotes*	Hershkovitz, 1985
	5	4	3·10	2·96	*C. s. satanus*	Hershkovitz, 1985; this study
Lagothrix flavicauda			≈10·0			Peres, 1994
Lagothrix lagotricha	16	9	7·28	7·02		T. Cole, pc
	4	3	>9·27	7·16	*L. l. cana* (two males weighed without digestive tract)	Peres, 1993; J. Kelley, pc
Pithecia aequatorialis			≈2·25			Emmons, 1990
Pithecia albicans	1		3·00			Ayres, 1986
Pithecia irrorata	2	2	2·25	2·07		Hershkovitz, 1987; Ferrari & Lopes, 1995
Pithecia monachus	16	10	2·61	2·11		Hernandez-Camacho & Defler, 1985; Ayres, 1986; Hershkovitz, 1987
Pithecia pithecia	10	4	1·94	1·58		Fooden, 1964, 1976*b*; Oliveira *et al.*, 1985; this study
Saimiri boliviensis	17	19	0·911	0·711		This study
Saimiri oerstedii	11	7	0·897	0·680		Crile & Quiring, 1940; Schultz, 1941
Saimiri sciureus	40	40	0·779	0·662	Colombia	Middleton & Rosal, 1972
	17	58	1·02	0·699	Monkey Jungle	Scollay, 1980
Saimiri ustus (incl. *madeirae*)	11	6	0·921	0·799	Mid-range of Ford & Davis	Ford & Davis, 1992; Ferrari & Lopes, 1995
Saimiri vanzolinii	9	4	0·950	0·650		Ayres, 1986
Cercopithecinae						
Allenopithecus nigroviridis	5	1	6·13	3·18		This study
	6		5·95	3·70	Captive and free-ranging	Gautier-Hion, 1988
Cercocebus agilis	2	2	9·50	5·66		Colyn, 1994
Cercocebus atys	3	4	11·0	6·2	*C. a. atys*	Oates *et al.*, 1990
Cercocebus galeritus	6	7	9·61	5·26		Gautier-Hion & Gautier, 1976; this study
Cercocebus torquatus	9	1	9·47	5·5		Sanders & Bodenbender, 1994; this study

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Cercopithecus aethiops	60	90	4·26	2·98	Kenya	Turner *et al.*, 1994
	29	30	5·51	4·09	*C. a. ngamiensis*, Botswana	Skinner & Smithers, 1990
	61	71	5·3	3·3	*C. a. sabaeus*, Barbados	Horrocks, 1986
Cercopithecus ascanius	69	242	3·70	2·92		Colyn, 1994
Cercopithecus campbelli	10	9	4·5	2·7	*C. c. campbelli*	Oates *et al.*, 1990
Cercopithecus cephus	15	10	4·29	2·88		Malbrant & Maclatchy, 1949; Gautier-Hion & Gautier, 1976
Cercopithecus diana	4	11	5·2	3·9	*C. d. diana*	Oates *et al.*, 1990
Cercopithecus dryas	1		3·0			Colyn *et al.*, 1991
Cercopithecus erythrogaster		1	≈4·1	2·4		Oates, 1985
Cercopithecus erythrotis	2		4·27			Napier, 1981
			≈3·6	≈2·9	Mid-range	Haltenorth & Diller, 1980
Cercopithecus hamlyni	11	9	5·49	3·36		Colyn, 1994
Cercopithecus lhoesti	19	50	5·97	3·45		Colyn, 1994
Cercopithecus mitis	33	34	7·93	4·25		Anapol *et al.*, 1995
	41	94	5·85	3·93	*C. m. stuhlmanni*	Colyn, 1994
	6	6	9·31	4·91	*C. m. erythrarchus*	Skinner & Smithers, 1990
Cercopithecus mona	2		5·1		*C. m. mona*	This study
Cercopithecus neglectus	6	6	7·35	4·13		Gautier-Hion & Gautier, 1976; Napier, 1981
		62		3·55		Gevaerts, 1992
Cercopithecus nictitans	33	30	6·67	4·26		Gautier-Hion & Gautier, 1976; Colyn, 1994
Cercopithecus petaurista	13	7	4·4	2·9	*C. p. buettikoferi*	Oates *et al.*, 1990
Cercopithecus pogonias	5	10	4·26	2·90		Gautier-Hion & Gautier, 1976; Colyn, 1994
Cercopithecus preussi			≈4·5			Juste *et al.*, 1995
Cercopithecus sclateri			≈4·0	≈2·5		Oates *et al.*, 1992
Cercopithecus solatus	4	3	6·89	3·92		Harrison, 1988; Gautier *et al.*, 1992
Cercopithecus wolfi	17	120	3·91	2·87		Colyn, 1994
Erythrocebus patas	9	14	12·4	6·5		Galat-Luong *et al.*, 1996
Lophocebus albigena	26	22	8·25	6·02		Gautier-Hion & Gautier, 1976; Napier, 1981; Colyn, 1994; this study
Lophocebus aterrimus	6	6	7·84	5·76		Colyn, 1994; this study
Macaca arctoides	7	3	12·2	8·4		Fooden, 1990
Macaca assamensis	21	15	11·3	6·9		Fooden, 1988
Macaca brunnescens	1		16·6		Captive (zoos and pets)	Watanabe *et al.*, 1987
Macaca cyclopis	7	4	6·00	4·94		Rothenfluh, 1976

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Macaca fascicularis	69	46	5·36	3·59		Fooden, 1995
Macaca fuscata	10	23	11·0	8·03		Kimura & Hanada, 1995
Macaca hecki	3	2	11·2	6·80	Captive (zoos and pets)	Hamada *et al.*, 1995
Macaca maura	17	4	9·72	6·05		Supriatna, 1991
Macaca mulatta	59	28	11·0	8·80	Cayo Santiago	Schwartz & Kemnitz, 1992
	8	9	7·71	5·37		Napier, 1981
Macaca nemestrina	8	10	11·2	6·5	*M. n. nemestrina*	Fooden, 1975
	6	7	7·7	4·9	*M. n. leonina*	
Macaca nigra	11	3	9·89	5·47		J. Supriatna & J. Froehlich, pc
Macaca nigrescens		1		9·5	Captive (zoos and pets)	Watanabe *et al.*, 1987
Macaca ochreata	2	1	5·30	2·60		This study
Macaca radiata	13	14	6·67	3·85		Fooden, 1981
Macaca silensis	11	11	8·9	6·1	San Diego Zoo	Harvey *et al.*, 1991
	1		6·75			Napier & Napier, 1967
Macaca sinica	23	55	5·68	3·20		Cheverud *et al.*, 1992
Macaca sylvanus			≈16·0	≈11·0	Provisioned wildlife parks	Fa, 1984
	1		11·1			Napier & Napier, 1967
Macaca thibetana	12	17	18·3	12·8	Provisioned by tourists	Zhao Qikum & Deng Ziyun, 1988
	5	1	15·2	9·5		Fooden *et al.*, 1985
Macaca tonkeana	6	3	14·9	9·0		Supriatna, 1991
Mandrillus leucophaeus	1		20·0			Malbrant & Maclatchy, 1949
			≈17·5	≈12·5	Mid-range	Haltenorth & Diller, 1980
Mandrillus sphinx	13	8	31·6	12·9	6M and 3F semi-free ranging	Malbrant & Maclatchy, 1949; Wickings & Dixson, 1992; Dixson *et al.*, 1993; this study
Miopithecus talapoin	7	9	1·38	1·12	Gabon	Gautier-Hion & Gautier, 1976
	1	2	2·5	2·0	Cameroon	This study
Papio anubis	100	116	25·1	13·3	Kenya	Popp, 1983; Gest & Siegel, 1983; Eley *et al.*, 1989
	39	35	21·2	11·7	Awash, Ethiopia	Phillips-Conroy & Jolly, 1981
Papio cynocephalus	37	21	21·8	12·3	Mikumi, Tanzania	J. Phillips-Conroy & J. Rogers, pc
	1	1	17·2	9·75	*P. c. kindae*	Napier, 1981
Papio hamadryas	41	39	16·9	9·9	Awash, Ethiopia	Phillips-Conroy & Jolly, 1981
	7	13	21·0	11·4	Harar, Ethiopia	Popp, 1983
Papio papio		3		12·1		Rothenfluh, 1976

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Papio ursinus	28	22	29·8	14·8	Botswana	Bulger & Hamilton, 1987; Skinner & Smithers, 1990
Theropithecus gelada	5	8	19·0	11·7		Dechow, 1983
Colobinae						
Colobus angolensis	12	11	9·68	7·57		Oates *et al.*, 1994; Colyn, 1994
Colobus guereza	3	4	13·5	9·2	*C. g. guereza*	Oates *et al.*, 1994
	13	14	9·89	7·90	*C. g. matschiei*	This study
Colobus polykomos	5	10	9·9	8·3	*C. p. polykomos*	Oates *et al.*, 1994
Colobus satanas	5	5	10·4	7·42		Malbrant & Maclatchy, 1949; Sabater Pi, 1973; Oates *et al.*, 1994
Colobus vellerosus	3	5	8·5	6·9		Oates *et al.*, 1994
Nasalis larvatus	46	48	20·4	9·82		Rothenfluh, 1976
Presbytis comata (=aygula)	4	6	6·68	6·71		Rothenfluh, 1976
Presbytis femoralis	23	18	6·26	6·19		Willis, 1995
Presbytis frontata	1	8	5·56	5·67		Willis, 1995
Presbytis hosei	7	3	6·18	5·63		Willis, 1995
Presbytis melalophos	11	12	6·59	6·47		Willis, 1995
Presbytis potenziani	8	4	6·17	6·40		Tilson & Tenaza, 1976; Brandon-Jones, 1993; Willis, 1995
Presbytis rubicunda	35	38	6·29	6·17		Willis, 1995
Presbytis thomasi	3	5	6·77	6·69		Willis, 1995
Procolobus badius	9	30	8·36	8·21	*P. b. badius*	This study
	1	3	5·80	5·46	*P. b. kirkii*	Napier, 1985
	3	7	9·67	7·21	*P. b. rufomitratus*	This study
	2	10	7·65	5·68	*P. b. langi*	Colyn, 1994
	1	6	9·20	7·47	*P. b. parmentierorum*	Colyn, 1994
	2	2	12·3	8·25	*P. b. oustaleti*	Colyn, 1994; this study
Procolobus verus	20	14	4·7	4·2		Oates *et al.*, 1990
Pygathrix nemaeus	4	2	11·0	8·44		Willis, 1995; this study
Pygathrix (Rhinopithecus) avunculus		1		8·00		Willis, 1995
	?	?	14·0	8·5		Boonratana & Le, 1997
Pygathrix (Rhinopithecus) bieti	1	5	≈15·0	9·96		Tan Bangjie, 1985; Jablonski & Pan Ruliang, 1995; Ji Weizhi *et al.*, 1994
Pygathrix (Rhinopithecus) brelichi	1		15·8			Tan Bangjie, 1985

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Pygathrix (Rhinopithecus) roxellana	7	4	17·9	11·6		Jablonski & Pan Ruliang, 1995
Semnopithecus entellus	5	9	19·2	14·8	*S. e. schistacea* group	
	9	11	13·0	9·89	*S. e. entellus* group	Napier, 1985; Willis, 1995
	14	11	11·4	6·91	*S. e. thersites* group	
Simias concolor	3	3	9·15	6·80		Napier, 1985; Willis, 1995
Trachypithecus auratus						
Trachypithecus cristatus	26	49	6·61	5·76		Willis, 1995
Trachypithecus francoisi	3	2	7·70	7·35		Willis, 1995
Trachypithecus geei	4	1	10·8	9·50		Oates *et al.*, 1994; Willis, 1995
Trachypithecus johnii	7	3	12·0	11·2		Willis, 1995
Trachypithecus obscurus	31	55	7·90	6·26		Burton, 1981; Willis, 1995
Trachypithecus phayrei	11	21	7·87	6·30		Napier, 1985; Willis, 1995
Trachypithecus pileatus	7	5	12·0	9·86		Willis, 1995
Trachypithecus vetulus (=senex)	3	3	8·17	5·90		Napier, 1985
Hylobatidae						
Hylobates agilis	19	10	5·88	5·82		Geissmann, 1993
Hylobates concolor	28	13	7·79	7·62		Ma Shilai *et al.*, 1988; Geissmann, 1993
Hylobates gabriellae						
Hylobates hoolock	13	5	6·87	6·88		Geissmann, 1993
Hylobates klossii	2	4	5·67	5·92		Jungers, 1984*b*
Hylobates lar	84	66	5·90	5·34		Geissmann, 1993
Hylobates leucogenys	8	4	7·41	7·32		Geissmann, 1993
Hylobates moloch	1	1	6·58	6·25		Geissmann, 1993
Hylobates muelleri	20	19	5·71	5·35		Geissmann, 1993
Hylobates pileatus	1	1	5·50	5·44		Geissmann, 1993
Hylobates syndactylus	7	10	11·9	10·7	Sumatra	Orgeldinger, 1994
Hominidae						
Gorilla gorilla	10	3	170·4	71·5	*G. g. gorilla* (Western lowland)	
	5	1	162·5	97·5	*G. g. beringei* (Mountain)	Jungers & Susman, 1984; C. Groves, pc
	4	2	175·2	71·0	*G. g. graueri* (Eastern lowland)	
Pan paniscus	7	6	45·0	33·2		Jungers & Susman, 1984
Pan troglodytes	21	26	42·7	33·7	*P. t. schweinfurthii*	Uehara & Nishida, 1987; this study
	5	4	59·7	45·8	*P. t. troglodytes*	Jungers & Susman, 1984; this study
	1	3	46·3	41·6	*P. t. verus*	This study

Table 5 ***Continued from previous page***

	Sample size M	Sample size F	Body mass, kg M	Body mass, kg F	Comments	References
Pongo pygmaeus	3	4	77·9	35·6	*P. p. abelli* (Sumatra)	Jungers, 1988, 1997; Markam & Groves, 1990
	7	13	78·5	35·8	*P. p. pygmaeus* (Borneo)	
Homo sapiens	405	319	47·9	42·2	Pygmies, Central African Republic	Pennetti *et al.*, 1986
	49	49	53·9	45·8	Highland subsistence farmers, Guatemala	Diaz *et al.*, 1991
	194	193	58·1	49·7	Melanesians, Bougainville	Friedlaender, 1987
	121	162	60·2	53·6	Australian aboriginals	Jones & White, 1994
	411	494	62·6	56·3	Saudi Arabians	Khalid, 1995
	62	55	65·0	52·5	Japanese	Abe *et al.*, 1994
	502	165	72·1	62·1	Danish	Holloway, 1980
	75	108	77·8	73·2	Western Samoans	Pawson, 1986

Abbreviations used in Comments column: pc, personal communication; AMNH, American Museum of Natural History; BMNH, British Museum of Natural History; FNMH, Field Museum of Natural History; MCZ, Museum of Comparative Zoology, Harvard University, NMNH, National Museum of Natural History, Smithsonian Institutions.

All data are for free-ranging animals unless specifically indicated otherwise in the Comments column.

The third data set often used in an interspecific comparative study concerns the body masses of the species, which are needed in order to control for or evaluate the pervasive effects of body size on variation in other traits. It is here where conventions have developed in primatology and physical anthropology that lead investigators to accept dated, undocumented, and just plain wrong data, potentially invalidating all their good work. For example, we think a case can be made that the data table published by Harvey & Clutton-Brock (1985) and Harvey *et al.* (1987) should *never* have been acceptable as a source. There are no references, no sample sizes, and values transparently incorrect to anyone who would give them a second look. Yet they have been used widely, and by meticulous workers in detailed studies in which it is evident that great effort was expended to obtain valid data and conduct correct analyses in all other parts of the project (e.g., Godfrey *et al.*, 1991; Aiello & Wood, 1994; Harcourt *et al.*, 1995).

By using excellent studies as examples, we are making the point that this issue involves a widely-held perspective on the strategy by which a comparative data set is put together, and on the significance of this step in a comparative study. It appears that it is often routine for investigators, referees, editors, and readers to quickly and superficially accept a table of data and to move on to emphasize the analyses and interpretations. But these latter steps are worthless if the data are wrong. We argue here for a cultural shift in our approach to data. Increased vigilance, skepticism, and more hard work are needed in the process of data collection. One certain consequence is that we would see smaller numbers of species in analyses because data based mostly on wishful thinking would be eliminated. Results reflect the data used, and we have evidence from the uncritical use of the data set from Harvey & Clutton-Brock (1985) that the standards applied to data collection have not routinely screened out clearly wrong data.

Excluding journals such as *Nature* and *Science*, in which space is not available, and excessively long tables of individual specimens, it follows that comparative studies published without the tables of data on which they are based (or complete citation of accessible references) are incomplete contributions. The work cannot be repeated or verified by others. Omitting the data implies that the data are not important or that nothing could possibly be disputed about them.

Finally, it is important to recognize that the best, carefully collected data for a species are often not good enough. For example, we would not place any credence in an estimate of sex dimorphism based on one or two specimens of each sex. Thus, even if the table presented in this study has succeeded in summarizing what is known about primate body mass, every body mass should not be used for every purpose.

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Appendix

Thirty-nine of the species and subspecies in Table 5 include data identified as collected for "this study." These previously unpublished data are masses taken from specimens in museums or private collections by W.L.J. The species are listed in the order that they appear in Table 5. The four values following the species name are, in order, number of males, number of females, mean body mass of males, and mean body mass of females. These data are followed by the name of the museum or collector holding the specimens, and by any additional information concerning the original geographic distribution of the specimens. All data are from wild-shot individuals. When the data for a species of subspecies have not been pooled with any other source in calculating the species means in Table 5, the values that can be directly read from Table 5 are not repeated below. Instead, the species name is followed immediately by the source of the data. See end of Table 5 for museum abbreviations.

Lepilemur edwardsi 0, 1, −, 0·925, H.-J. Kuhn collection (Gottingen)
Propithecus diadema 0, 2, −, 6·6, NMNH
Arctocebus calabarensis 1, 1, 0·300, 0·346, NMNH, from Nigeria
Nycticebus coucang bengalensis 2, 1, 1·31, 1·14, MCZ and BMNH
Perodicticus potto edwardsi 1, 1, 1·36, 1·40, AMNH and Powell-Cotton
Perodicticus potto potto BMNH, NMNH
Callithrix penicillata NMNH, all from Brazil
Saguinus midas 8, 6, 0·544, 0·601, NMNH, all from Brazil. Also: 11, 7, 0·526, 0·580, Leiden, all from Surinam
Alouatta fusca NMNH, all from Brazil
Aotus azarae AMNH, most from Bolivia
Aotus trivirgatus NMNH, all from Venezuela
Ateles belzebuth 10, 16, 8·26, 7·88, NMNH, all from Venezuela
Callicebus donacophilus AMNH, all but one from Bolivia
Callicebus hoffmannsi 1, 3, 1·09, 1·07, NMNH, all from Brazil
Callicebus torquatus 8, 6, 1·15, 1·05, NMNH, all from Venezuela
Cebus albifrons 18, 15, 3·07, 2·18, AMNH from Bolivia and NMNH from Venezuela
Cebus apella 13, 10, 3·84, 2·66, Leiden, all from Surinam
Cebus olivaceus 14, 7, 3·38, 2·79, NMNH (N=20), FMNH (N=1), most from Venezuela

Chiropotes satanus satanus 4, 3, 3·25, 3·11, Leiden, all from Surinam
Pithecia pithecia 2, 0, 1·94, −, Leiden, from Surinam
Saimiri boliviensis AMNH and NMNH
Allenopithecus nigroviridis Tervuren and R. Susman collection (Stony Brook)
Cercocebus galeritus 3, 4, 9·03, 5·10, Tervuren
Cercocebus torquatus 6, 0, 10·2, −, Powell-Cotton
Cercopithecus mona Powell-Cotton
Lophocebus albigena 9, 3, 8·92, 6·05, Tervuren and R. Susman collection (Stony Brook)
Lophocebus aterrimus 5, 2, 7·83, 6·00, Tervuren and R. Susman collection (Stony Brook)
Macaca ochreata NMNH
Mandrillus sphinx 1, 3, 45·0, 13·0, Powell-Cotton
Miopithecus talapoin Powell-Cotton
Colobus guereza matschiei NMNH, BMNH, and Powell-Cotton
Procolobus badius badius H.-J. Kuhn collection (Gottingen)
Procolobus badius rufomitratus Tervuren
Procolobus badius oustaleti 1, 0, 12·1, −, Tervuren
Pygathrix nemaeus 2, 1, 11·0, 8·18, NMNH
Pan troglodytes schweinfurthii 3, 3, 53·7, 35·9, NMNH and field notes (Vandebroek)
Pan troglodytes troglodytes 2, 1, 59·3, 41·0, AMNH and NMNH
Pan troglodytes verus NMNH and R. Susman collection (Stony Brook)